UE26CY151A · ENGINEERING CHEMISTRY

SEMESTER 1 · UNIT 1

01

FORMULA SHEET

Formulas & Symbols

Every formula in Unit 1 as a card with conditions and units, followed by a full glossary of the symbols they use.

M1
Molecular Spectroscopy
8 topics

M2
Phase Equilibria
5 topics

M3
Computational Chemistry
1 topic

Exam weight	★★★ high · ★★☆ medium · ★☆☆ low
Flags	NOT ON CURRENT SLIDES · TRANSCRIPT ONLY · ADDED FOR CLARITY
Sources	S004·8 slide 8 of deck 004 · T009 class transcript · PDF Unit-1 slide PDF · ESA past paper

Compiled from PESU AV-summary transcripts, class slides and past ESA papers · 13 October 2026

Contents

M1 Molecular Spectroscopy

T1 EM radiation & interaction with matter ★★★

T2 Electromagnetic spectrum & regions ★★★

T3 Born–Oppenheimer approximation ★★☆

T3b Beer–Lambert law ★☆☆

T4 Rotational (microwave) spectroscopy ★★★

T5 Vibrational (IR) – SHO ★★★

T6 Vibrational – anharmonic ★★☆

T7 Electronic spectroscopy & Franck–Condon ★★★

M2 Phase Equilibria

T8 Basic terms & phase rule ★★★

T9 Derivation of phase rule ★★★

T10 One-component system: water ★★★

T11 Two-component: reduced phase rule & Pb–Ag ★★★

T12 Thermal analysis / cooling curves ★★☆

M3 Computational Chemistry

T13 Computational chemistry ★☆☆

+ Constants & unit conversions

+ Symbol glossary

MODULE 1

Molecular Spectroscopy

T1 · EM radiation & interaction with matter | T2 · Electromagnetic spectrum & regions | T3 · Born–Oppenheimer approximation | T3b · Beer–Lambert law | T4 · Rotational (microwave) spectroscopy | T5 · Vibrational (IR) – SHO | T6 · Vibrational – anharmonic | T7 · Electronic spectroscopy & Franck–Condon

T1 EM radiation & interaction with matter ★★★

F1 Photon energy ★★★

$$E = h\nu$$

GIVES Energy of radiation
APPLIES any EM radiation
UNITS J

S001·5

F2 Frequency–wavelength ★★★

$$\nu = \frac{c}{\lambda}$$

GIVES Frequency
APPLIES vacuum/air
UNITS Hz

S001·5

F3 Wavenumber ★★★

$$\bar{\nu} = \frac{1}{\lambda} \qquad E = hc\bar{\nu}$$

GIVES Wavenumber; energy
APPLIES spectroscopic unit
UNITS cm^{-1}

S001·5

F4 Transition energy ★★★

$$\Delta E = E_2 - E_1 = h\nu$$

GIVES Absorbed/emitted quantum
APPLIES two quantised levels
UNITS J (÷hc → cm^{-1})

S001

T2 Electromagnetic spectrum & regions ★★★

No formula in this topic — see the important points.

T3 Born–Oppenheimer approximation ★★☆

F5 Total molecular energy ★★☆

$$E = E_{trans} + E_{rot} + E_{vib} + E_{elec} \approx E_{rot} + E_{vib} + E_{elec}$$

GIVES Born–Oppenheimer
APPLIES E_{trans} negligible
UNITS J

S003·4

F6 **Ordering** ★★☆

$$E_{rot} \ll E_{vib} \ll E_{elec}$$

GIVES Microwave < IR < UV-Vis
UNITS —

S003·4

F7 **Change in energy** ★★☆

$$\Delta E_{total} = \Delta E_{elec} + \Delta E_{vib} + \Delta E_{rot}$$

GIVES Electronic transition fine structure
UNITS J or cm^{-1}

S007·4

T3b Beer–Lambert law ★☆☆

F8 **Beer–Lambert** ★☆☆

$$A = \log \frac{I_0}{I} = \varepsilon c l$$

NOT ON CURRENT SLIDES
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GIVES Absorbance
APPLIES dilute solutions
UNITS dimensionless

T4 Rotational (microwave) spectroscopy ★★★

F9 **Centre of mass** ★★★

$$m_1 r_1 = m_2 r_2 \qquad r_0 = r_1 + r_2$$

GIVES Positions of atoms about C
APPLIES rigid diatomic
UNITS m

S004; PDF

F10 **Distances** ★★★

$$r_1 = \frac{m_2\, r_0}{m_1 + m_2} \qquad r_2 = \frac{m_1\, r_0}{m_1 + m_2}$$

UNITS m

PDF

F11 **Moment of inertia** ★★★

$$I = m_1 r_1^2 + m_2 r_2^2 = \mu r_0^2, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

GIVES Moment of inertia
APPLIES rigid rotor
UNITS kg m^2

S004·7; PDF

F12 Rotational energy ★★☆

$$E_r = \tfrac{1}{2} I\omega^2 = \frac{L^2}{2I}$$

GIVES Classical rotor
UNITS J

PDF

F13 Rotational levels (J) ★★★

$$E_J = \frac{h^2}{8\pi^2 I}\, J(J+1)\ \mathrm{J}$$

GIVES Quantised levels
APPLIES J = 0,1,2…
UNITS J

S004·8

F14 Rotational levels (cm⁻¹) ★★★

$$\varepsilon_J = \frac{h}{8\pi^2 Ic}\, J(J+1) = BJ(J+1)\ \mathrm{cm}^{-1}, \qquad B = \frac{h}{8\pi^2 Ic}$$

GIVES Level energy; rotational constant
APPLIES c in cm s^{-1}
UNITS cm^{-1}

S004·8

F15 Selection rules ★★★

$$\Delta J = \pm 1$$

GIVES Allowed transitions
APPLIES gross rule
UNITS —

S004·10

F16 Line positions ★★★

$$\Delta\varepsilon_{J\to J+1} = 2B(J+1)\ \mathrm{cm}^{-1}; \quad \text{levels } 0, 2B, 6B, 12B, 20B, 30B, 42B$$

GIVES Rotational spectrum
UNITS cm^{-1}

S004·9-10

T5 Vibrational (IR) – SHO ★★★

F17 Hooke's law ★★☆

$$F = -kx \qquad PE = \tfrac{1}{2}k(r - r_{eq})^2$$

GIVES Parabolic potential
APPLIES SHO
UNITS N ; J

S005·6-7

F18 Oscillation frequency ★★★

$$\nu_{osc} = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}}\ \mathrm{Hz}$$

GIVES Frequency
APPLIES SHO
UNITS Hz

S005·8

F19 **Wavenumber of vibration** ★★★

$$\bar{\nu}_{osc} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}\ \mathrm{cm}^{-1}$$

GIVES Line position
APPLIES c in cm s^{-1}
UNITS cm^{-1}

S005·8

F20 **SHO levels** ★★★

$$E_v = \left(v + \tfrac{1}{2}\right) h\nu_{osc}\ \mathrm{J} \qquad \varepsilon_v = \left(v + \tfrac{1}{2}\right) \bar{\nu}_{osc}\ \mathrm{cm}^{-1}$$

GIVES Levels
APPLIES v = 0,1,2…
UNITS J ; cm^{-1}

S005·8-9

F21 **Zero-point energy** ★★★

$$\varepsilon_0 = \tfrac{1}{2}\,\bar{\nu}_{osc}$$

GIVES Lowest level (≠ 0)
UNITS cm^{-1}

S005·9

F22 **Selection rule & spacing** ★★★

$$\Delta v = \pm 1; \quad \varepsilon_{v+1} - \varepsilon_v = \bar{\nu}_{osc} \ \Rightarrow\ \text{one line at } \bar{\nu}_{osc}$$

APPLIES dipole must change
UNITS cm^{-1}

S005·9

F23 **Force constant** ★★★

$$k = 4\pi^2 c^2 \bar{\nu}^2 \mu$$

GIVES Bond stiffness
APPLIES μ in kg
UNITS N m^{-1}

derived

T6 Vibrational – anharmonic

★★☆

F24 Morse function

★★☆

$$PE = D_{eq}\left[1 - \exp\{a(r_{eq} - r)\}\right]^2$$

GIVES Anharmonic potential
UNITS J

S006·3

F25 Anharmonic levels

★★☆

$$\varepsilon_v = \left(v + \tfrac{1}{2}\right)\bar{\omega}_e - \left(v + \tfrac{1}{2}\right)^2 \bar{\omega}_e x_e \ \text{cm}^{-1}$$

GIVES Converging levels
APPLIES x_e small, positive
UNITS cm^{-1}

S006·4

F26 Transitions

★★☆

$$\begin{aligned} 0 \to 1 &: \ \bar{\omega}_e(1 - 2x_e) \quad \text{fundamental} \\ 0 \to 2 &: \ 2\bar{\omega}_e(1 - 3x_e) \quad \text{1st overtone} \\ 0 \to 3 &: \ 3\bar{\omega}_e(1 - 4x_e) \quad \text{2nd overtone} \end{aligned}$$

APPLIES Δv = ±1,±2,±3…
UNITS cm^{-1}

S006·6-7

F27 Hot band & ZPE

★★☆

$$1 \to 2 : \ \bar{\omega}_e(1 - 4x_e) \qquad \varepsilon_0 = \tfrac{1}{2}\bar{\omega}_e\left(1 - \tfrac{1}{2}x_e\right)$$

ADDED FOR CLARITY

APPLIES higher T
UNITS cm^{-1}

S006·9; ZPE added

T7 Electronic spectroscopy & Franck–Condon

★★★

F28 Electronic spectrum

★★☆

$$\varepsilon = \varepsilon_{elec} + \varepsilon_{vib}; \quad \text{lines labelled } (v', v'')$$

GIVES Progression
APPLIES no vibrational selection rule
UNITS cm^{-1}

S007·6-8

MODULE 2

Phase Equilibria

T8 · Basic terms & phase rule | T9 · Derivation of phase rule | T10 · One-component system: water | T11 · Two-component: reduced phase rule & Pb–Ag | T12 · Thermal analysis / cooling curves

T8 Basic terms & phase rule ★★★

F29 Phase rule ★★★

$$F = C - P + 2$$

GIVES Degrees of freedom
APPLIES no external fields
UNITS —

S008·12

T9 Derivation of phase rule ★★★

F30 Variables & constraints ★★★

$$F = \underbrace{P(C-1)+2}_{\text{variables}} - \underbrace{C(P-1)}_{\text{constraints}} = C - P + 2$$

GIVES Derivation
UNITS —

S009·7

F31 Equilibrium condition ★★★

$$\mu_i^{\alpha} = \mu_i^{\beta} = \mu_i^{\gamma} = \cdots$$

GIVES Equal chemical potential
APPLIES each component
UNITS J mol^{-1}

S009·3,7

F32 Chemical potential ★★☆

$$\mu_i = \left(\frac{\partial G}{\partial n_i}\right)_{T,P,n_j}$$

TRANSCRIPT ONLY

GIVES Partial molar Gibbs energy
UNITS J mol^{-1}

T008 (S009·5 unreadable)

T10 One-component system: water ★★★

F33 One-component ★★★

$$F = 3 - P$$

GIVES Area F=2, line F=1, point F=0
UNITS —

S009·8

F34 Clausius–Clapeyron ★★★

$$\frac{dP}{dT} = \frac{\Delta H}{T\,\Delta V}$$

GIVES Slope of curve
APPLIES fusion, vaporisation, sublimation
UNITS Pa K^{-1}

S009·12-14

T11 Two-component: reduced phase rule & Pb–Ag ★★★

F35 Reduced (condensed) rule ★★★

$$F = C - P + 1$$

GIVES Constant pressure
APPLIES condensed systems
UNITS —

S010·2-3

T12 Thermal analysis / cooling curves ★★☆

No formula in this topic — see the important points.

MODULE 3

Computational Chemistry

T13 · Computational chemistry

T13 Computational chemistry

★☆☆

No formula in this topic — see the important points.

Constants & unit conversions

h	6.626×10^{-34} J s	Planck constant · slides also use 6.625
c	3×10^{8} m s^{-1} = 3×10^{10} cm s^{-1}	speed of light · use cm s^{-1} with cm^{-1}
N_A	6.022×10^{23} mol^{-1}	Avogadro constant
amu	1.66×10^{-27} kg	atomic mass unit · μ must be in kg
π	3.14	pi · as used in papers

- J → cm^{-1} **divide by hc**
- cm^{-1} → Hz **multiply by c**
- 1 eV **1.602×10^{-19} J = 8065.5 cm^{-1}**
- 1 kJ mol^{-1} **83.59 cm^{-1}**
- 1 Å **10^{-10} m**

Symbol glossary

Same letter, different meaning. μ is reduced mass (spectroscopy) and chemical potential (phase rule). P is number of phases and pressure. ε is energy in cm^{-1} and molar absorptivity (Beer–Lambert).

Symbol	Meaning in this unit	Unit	Value / note
λ	wavelength	m, nm, Å	
ν	frequency	Hz	
$\bar{\nu}$	wavenumber	cm^{-1}	
c	speed of light	$m\ s^{-1}$ / $cm\ s^{-1}$	3×10^{8} $m\ s^{-1}$ = 3×10^{10} $cm\ s^{-1}$
h	Planck constant	J s	6.626×10^{-34} (slides also use 6.625)
E, ε	energy; ε = energy in cm^{-1} (also molar absorptivity in Beer–Lambert — 3rd meaning!)	J; cm^{-1}; L mol^{-1} cm^{-1}	
ΔE	energy change	J	
J	rotational quantum number	—	0,1,2…
B	rotational constant = $h/8\pi^2Ic$	cm^{-1}	
I	moment of inertia	kg m^2	
μ	reduced mass $m_1m_2/(m_1+m_2)$ — ALSO chemical potential in T8–T9	kg; J mol^{-1}	1 amu = 1.66×10^{-27} kg
m_1, m_2	atomic masses	kg	
r_0	bond length (rigid rotor)	m, Å	1 Å = 10^{-10} m
r_1, r_2	distances of atoms from centre of mass	m	
ω	angular velocity	rad s^{-1}	
L	angular momentum	kg m^2 s^{-1}	
k	force constant	N m^{-1}	
x	displacement	m	
v	vibrational quantum number	—	0,1,2…
ν_{osc}, $\bar{\nu}_{osc}$	oscillation frequency / wavenumber	Hz ; cm^{-1}	
$\bar{\omega_e}$	equilibrium vibrational wavenumber (anharmonic)	cm^{-1}	
x_e	anharmonicity constant	—	small, positive
D_{eq}	depth of Morse well	J	

Symbol	Meaning in this unit	Unit	Value / note
a	Morse constant	m^{-1}	
r_{eq}	equilibrium bond length	m	
r_e', r_e''	bond length in excited / ground state	Å	
v', v''	vibrational number in excited / ground state	—	
$\varepsilon', \varepsilon''$	energy in excited / ground state	cm^{-1}	
F	degrees of freedom (variance)	—	
C	number of components	—	
P	number of phases — ALSO pressure (clash!)	— ; atm, mmHg	
T	temperature	K, °C	
G	Gibbs energy	J	
n_i	moles of component i	mol	
ΔH	enthalpy of transition	J mol^{-1}	
ΔV	volume change of transition	m^3 mol^{-1}	
A	absorbance	—	
I_0, I	incident / transmitted intensity	—	
c, l	concentration, path length (Beer–Lambert)	mol L^{-1}, cm	
N_A	Avogadro constant	mol^{-1}	6.022×10^{23}